

Improving Probabilistic Diffusion Models with Optimal Diagonal Covariance Matching

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01 Paper Title and Authors



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- We highlight **optimal covariance** and diffusion improvement
- Figure: we show sample generations from datasets used.

02 Motivation & Background (Not Method): Covariance in DDPM/DDIM

Modeling Capability	Covariance Type	+ #Passes	Intuition
	x_t -independent Isotropic β (Ho et al., 2020)	0	Cov. of $q(x_t x_{t-1})$
	x_t -independent Isotropic $\tilde{\beta}$ (Ho et al., 2020)	0	Cov. of $q(x_t x_{t-1}, x_0)$
	x_t -independent Isotropic Estimation (Bao et al., 2022b)	0	Estimate from data
	x_t -dependent Diagonal VLB (Nichol & Dhariwal, 2021)	1	Learn from data
	x_t -dependent Diagonal NS (Bao et al., 2022a)	1	Learn from data
	x_t -dependent Diagonal OCM (Ours)	1	Learn from score
	x_t -dependent Diagonal Estimation (Zhang et al., 2024b)	$2M$	Estimate from score
	x_t -dependent Diagonal Analytic (Zhang et al., 2024b)	D	Calculate from score
	x_t -dependent Full Analytic (Zhang et al., 2024b)	D	Calculate from score

- We show heuristic/diagonal covariances hurt when steps are few
- We explain why covariance matters most at small T: error compounds
- Table: we overview choices; takeaway—learned, state-dependent helps

03 Method: Our Proposal — Unbiased Optimal Covariance Matching (OCM)

# timesteps	FiD ↓				NLL ↓			
	5	10	15	20	5	10	15	20
DDPM, $\tilde{\beta}$	58.28	34.76	24.02	19.00	203.29	74.95	44.94	32.20
DDPM, β	254.07	205.31	149.67	109.81	7.33	6.51	6.06	5.77
OCM-DDPM	38.88	21.60	13.35	9.75	6.82	4.98	4.62	4.43

- We propose OCM: learn optimal diagonal covariance from the score
- We improve FID and NLL with negligible overhead across samplers
- Table: method scope; code: https://github.com/J-zin/OCM_DPM

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Method: Score-to-Covariance Identity — Intuition and Implications

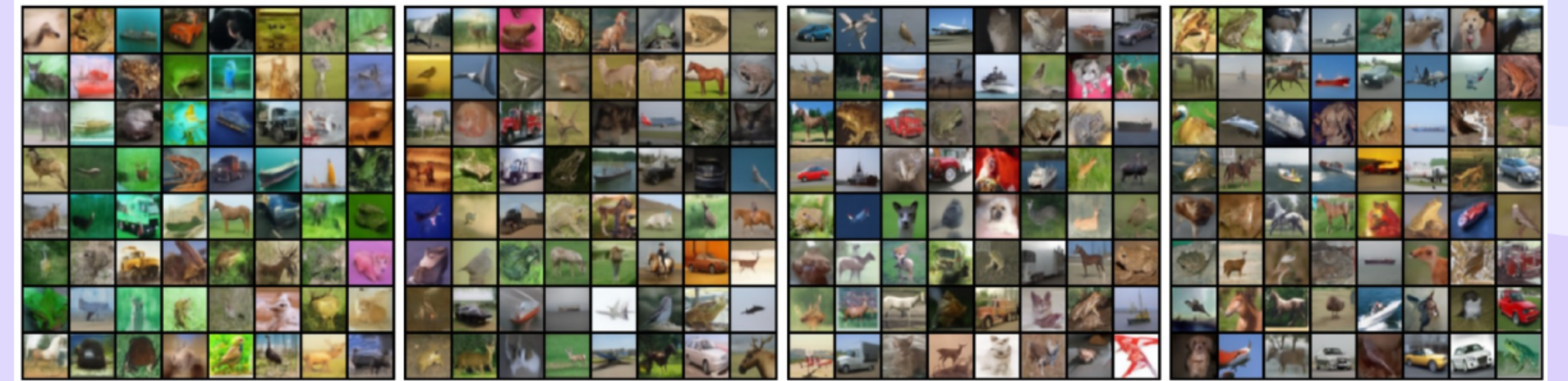
- We derive a score-to-covariance identity for scaled convolutions
- We note Hessians make exact evaluation memory/computation heavy
- Figure: identity table—covariance computed solely from the score.

# TIMESTEPS K	CIFAR10 (LS)						CIFAR10 (CS)					
	10	25	50	100	200	1000	10	25	50	100	200	1000
MEAN-DDPM (NLL)	5.33	4.63	4.35	3.97	3.78	3.57	4.99	4.34	3.99	3.76	3.59	3.41
STD-DDPM (NLL) %	0.04	0.03	0.03	0.04	0.04	0.01	0.02	0.01	0.03	0.03	0.02	0.01
MEAN-DDPM (FID)	25.07	9.30	5.90	4.37	3.54	4.00	14.46	5.49	4.09	3.83	3.81	4.22
STD-DDPM (FID) %	9.05	7.54	3.48	2.77	4.25	2.30	10.40	7.74	2.66	1.32	5.26	3.08
MEAN-DDIM (FID)	10.61	4.31	3.48	3.26	3.25	3.75	16.85	6.70	4.73	4.33	4.56	4.59
STD-DDIM (FID) %	5.17	3.55	3.19	1.87	3.97	0.99	13.22	4.23	2.28	2.94	1.61	5.30

# TIMESTEPS K	CELEBA 64X64						IMAGENET 64X64					
	10	25	50	100	200	1000	25	50	100	200	400	4000
MEAN-DDPM (NLL)	4.69	3.86	3.43	3.13	2.90	2.66	4.45	4.15	3.93	3.79	3.70	3.59
STD-DDPM (NLL) %	0.01	0.01	0.01	0.00	0.01	0.00	0.01	0.00	0.01	0.00	0.01	0.01
MEAN-DDPM (FID)	21.58	12.65	9.24	6.96	6.00	5.02	28.01	20.85	18.01	16.76	16.35	16.33
STD-DDPM (FID) %	2.24	7.65	2.29	2.32	5.50	1.37	0.77	4.66	6.82	3.28	3.04	3.21
MEAN-DDIM (FID)	10.36	5.69	4.37	3.50	3.11	3.03	28.34	19.68	17.84	17.41	17.36	18.91
STD-DDIM (FID) %	7.40	4.04	5.12	4.88	6.46	1.55	6.59	3.77	9.15	4.08	7.88	5.26

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Method: Practical Diagonal Estimation — Hessian Cost and Hutchinson Limits



(a) OCM-DDPM ($M=5$) (b) OCM-DDPM ($M=10$) (c) OCM-DDPM ($M=15$) (d) OCM-DDPM ($M=20$)

- We review Hutchinson diagonal estimation needing many samples
- We show compute scales with M ; sampling becomes impractical
- Figure: we see variance drops slowly; $M \geq 100$ often needed

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Method: OCM Objective — Why Unbiased, Why Fast

$K =$	FID ↓				NLL (%) ↑			
	10	25	50	100	10	25	50	100
$M=1$	14.32	5.54	4.10	3.84	4.99	4.34	3.99	3.76
$M=3$	14.18	5.51	4.11	3.82	4.99	4.34	3.99	3.76
$M=5$	14.17	5.51	4.11	3.82	4.99	4.34	3.99	3.76
$M=10$	14.16	5.51	4.11	3.82	4.99	4.34	3.99	3.76

- We define OCM: match expected $\text{diag}(H)$ without inner expectation
- We minimize unbiased MSE between $h \backslash \phi(x)$ and $\text{diag}(H(x))$.
- Table: OCM improves FID/NLL; takeaway— $M=1$ often suffices

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Method: Integration — Training/Inference in DDPM/DDIM

	NN_1	NN_2
CIFAR10 (LS)	Conv	Conv
CIFAR10 (CS)	Conv	Conv
CELEBA 64X64	Conv	Conv
IMAGENET 64X64	Conv	Res+Conv
LSUN BEDROOM	Conv	Res+Conv
IMAGENET 256X256	AdaLN+Linear	AdaLN+Linear

- We share a backbone; add a tiny NN_2 covariance head
- We state NN_2 adds negligible inference cost
- Table: datasets/backbones; NN_2 =covariance head; overhead negligible.

Method: Positioning vs. Baselines — beta/beta-tilde, A-, I-, SN/NPR; DDIM applicability and OCM $\leftrightarrow$ SN link

Covariance Type	+ #Passes	Intuition
x_t -independent Isotropic β (Ho et al., 2020)	0	Cov. of $q(x_t x_{t-1})$
x_t -independent Isotropic $\tilde{\beta}$ (Ho et al., 2020)	0	Cov. of $q(x_t x_{t-1}, x_0)$
x_t -independent Isotropic Estimation (Bao et al., 2022b)	0	Estimate from data
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x_t -dependent Diagonal OCM (Ours)	1	Learn from score
x_t -dependent Diagonal Estimation (Zhang et al., 2024b)	$2M$	Estimate from score
x_t -dependent Diagonal Analytic (Zhang et al., 2024b)	D	Calculate from score
x_t -dependent Full Analytic (Zhang et al., 2024b)	D	Calculate from score

- We position OCM among β , $\tilde{\beta}$, A-, I-, SN/NPR
- We learn covariance from the score; works for DDPM/DDIM
- Table: we show higher capacity at one-pass cost; broad applicability

Toy Validation: Known Scores and Optimal Covariance

- We validate on 2D mixtures and image generation
- We outperform baselines on MMD across steps
- Figure: we show lowest curves; full>diag, ours best diag

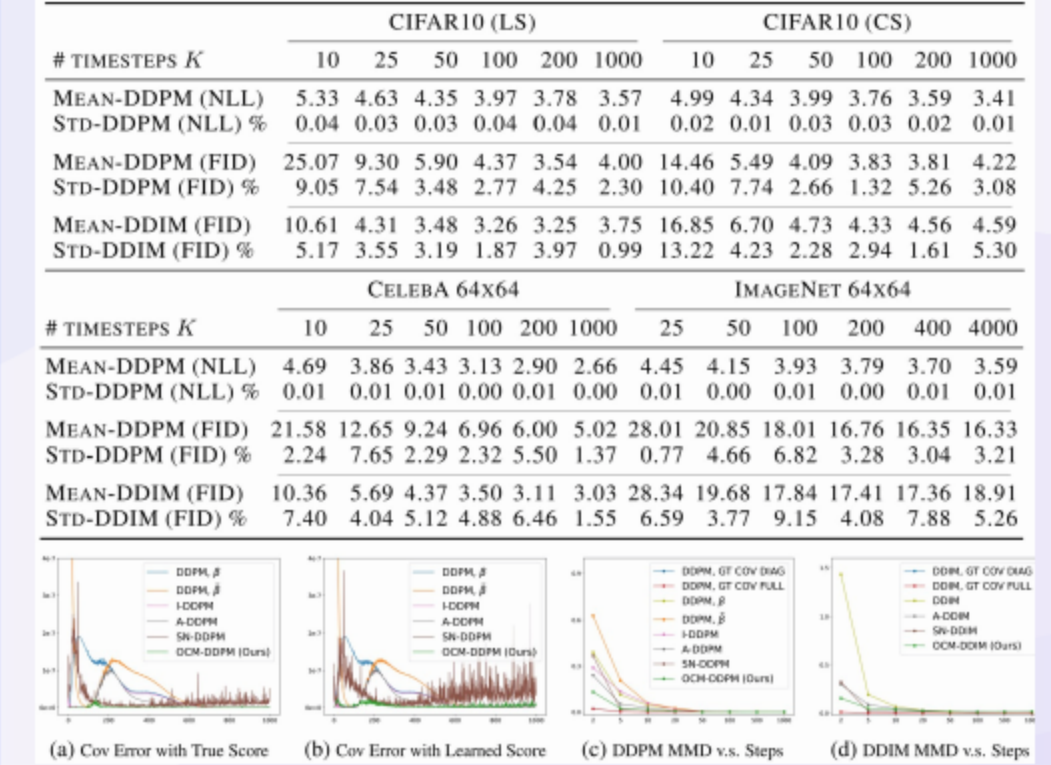
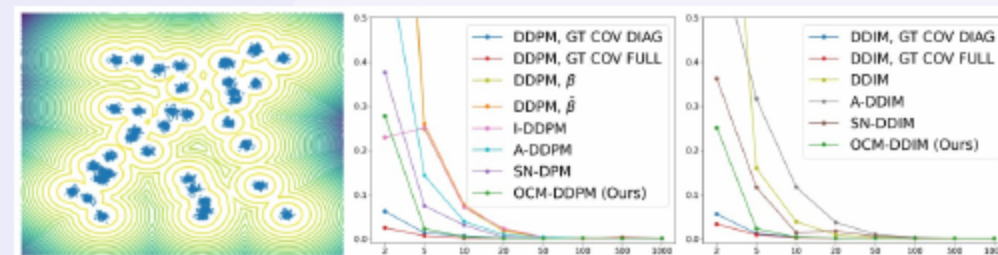


Image Results & Efficiency: FID-NLL Trade-offs and Steps

- We show FID-NLL tradeoffs; our curves Pareto-dominate.
- Table: we list fewest steps to FID $\approx$ 6 after overhead



METHOD	CIFAR10	CELEBA 64x64	LSUN BEDROOM	IMAGENET 256x256
DDPM	90 (6.12)	> 200	130 (6.06)	21 (5.89)
DDIM	30 (5.85)	> 100	BEST FID > 6	11 (5.58)
IMPROVED DDPM	45 (5.96)	MISSING MODEL	90 (6.02)	22 (6.08)
ANALYTIC-DPM	25 (5.81)	55 (5.98)	100 (6.05)	MISSING MODEL
NPR-DPM	1.002 $\times$ 23 (5.76)	1.013 $\times$ 50 (6.04)	1.021 $\times$ 90 (6.01)	MISSING MODEL
SN-DPM	1.005 $\times$ 17 (5.81)	1.019 $\times$ 22 (5.96)	1.114 $\times$ 92 (6.02)	MISSING MODEL
OCM-DPM (OURS)	1.003 $\times$ 16 (5.83)	1.015 $\times$ 21 (5.94)	1.112 $\times$ 90 (6.04)	1.007 $\times$ 10 (5.33)

Latent Diffusion and CFG: Quality, Diversity, and Steps

- We test latent DiT with CFG; best FID at CFG=2.0.
- Table: we show step counts; quality and recall improve



CFG =	FID $\downarrow$					RECALL (%) $\uparrow$				
	1.5	1.75	2.0	3.0	4.0	1.5	1.75	2.0	3.0	4.0
DDPM, $\tilde{\beta}$	30.41	19.26	13.53	11.52	14.76	39.74	35.29	32.08	24.93	19.91
DDPM, β	210.28	182.78	158.16	89.86	58.16	16.98	22.28	25.13	23.07	19.48
I-DDPM	44.96	20.71	13.01	9.76	13.75	50.32	43.75	41.08	31.77	25.49
OCM-DDPM	30.55	17.57	11.12	9.52	13.74	49.39	45.87	42.33	32.13	24.96
DDIM	9.41	6.54	6.12	10.49	14.18	49.15	45.82	41.15	29.27	23.16
OCM-DDIM	11.30	6.67	5.33	9.20	13.49	54.50	50.32	46.58	34.53	25.67

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Most Surprising: One-sample OCM matches **100**-sample MC at a fraction of compute.

$K =$	FID ↓				NLL (%) ↑			
	10	25	50	100	10	25	50	100
$M = 1$	14.32	5.54	4.10	3.84	4.99	4.34	3.99	3.76
$M = 3$	14.18	5.51	4.11	3.82	4.99	4.34	3.99	3.76
$M = 5$	14.17	5.51	4.11	3.82	4.99	4.34	3.99	3.76
$M = 10$	14.16	5.51	4.11	3.82	4.99	4.34	3.99	3.76

- We match 100-sample MC with a single OCM sample.
- Table: M=1 matches M=10 across K for FID/NLL.
- We boost diversity under CFG without FID loss

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Discussion & Conclusion: Limits, Future Directions, Key Takeaways

# Timesteps K	CIFAR10 (LS)						CIFAR10 (CS)					
	10	25	50	100	200	1000	10	25	50	100	200	1000
DDPM, β	44.45	21.83	15.21	10.94	8.23	5.11	34.76	16.18	11.11	8.38	6.66	4.92
DDPM, β	233.41	125.05	66.28	31.36	12.96	3.04	205.31	84.71	37.35	14.81	5.74	3.34
A-DDPM	34.26	11.60	7.25	5.40	4.01	4.03	22.94	8.50	5.50	4.45	4.04	4.31
NPR-DDPM	32.35	10.55	6.18	4.52	3.57	4.10	19.94	7.99	5.31	4.52	4.10	4.27
SN-DDPM	24.06	6.91	4.63	3.67	3.31	3.65	16.33	6.05	4.17	3.83	3.72	4.07
OCM-DDPM	<u>24.94</u>	<u>9.19</u>	<u>5.95</u>	<u>4.36</u>	<u>3.48</u>	<u>3.98</u>	14.32	5.54	4.10	<u>3.84</u>	<u>3.75</u>	<u>4.18</u>
DDIM	21.31	10.70	7.74	6.08	5.07	4.13	34.34	16.68	10.48	7.94	6.69	4.89
A-DDIM	14.00	5.81	4.04	3.55	3.39	3.74	26.43	9.96	6.02	4.88	4.92	4.66
NPR-DDIM	13.34	5.38	3.95	3.53	3.42	3.72	22.81	9.47	6.04	5.02	5.06	4.62
SN-DDIM	12.19	4.28	3.39	3.23	3.22	3.65	17.90	7.36	5.16	4.63	4.63	4.51
OCM-DDIM	10.66	<u>4.35</u>	<u>3.48</u>	<u>3.27</u>	<u>3.29</u>	<u>3.74</u>	16.70	6.71	4.72	4.30	4.54	<u>4.53</u>

# Timesteps K	CELEBA 64x64						IMAGENET 64x64					
	10	25	50	100	200	1000	25	50	100	200	400	4000
DDPM, β	36.69	24.46	18.96	14.31	10.48	5.95	29.21	21.71	19.12	17.81	17.48	16.55
DDPM, β	294.79	115.69	53.39	25.65	9.72	3.16	170.28	83.86	45.04	28.39	21.38	16.38
A-DDPM	28.99	16.01	11.23	8.08	6.51	5.21	32.56	22.45	18.80	17.16	16.40	16.34
NPR-DDPM	28.37	15.74	10.89	8.23	7.03	5.33	28.27	20.89	18.06	16.96	16.32	16.38
SN-DDPM	20.60	12.00	7.88	5.89	5.02	4.42	27.58	20.74	18.04	16.61	16.37	16.22
OCM-DDPM	<u>21.55</u>	<u>12.71</u>	<u>9.24</u>	<u>6.97</u>	<u>5.92</u>	<u>5.04</u>	<u>28.02</u>	<u>20.81</u>	17.98	<u>16.74</u>	16.32	<u>16.31</u>
DDIM	20.54	13.45	9.33	6.60	4.96	3.40	<u>26.06</u>	20.10	18.09	17.84	17.74	19.00
A-DDIM	15.62	9.22	6.13	4.29	3.46	3.13	25.98	19.23	17.73	17.49	17.44	18.98
NPR-DDIM	14.98	8.93	6.04	4.27	3.59	3.15	28.84	19.62	<u>17.63</u>	<u>17.42</u>	17.30	18.91
SN-DDIM	10.20	5.48	3.83	3.04	2.85	2.90	28.07	19.38	17.53	17.23	17.23	18.89
OCM-DDIM	<u>10.28</u>	<u>5.72</u>	<u>4.42</u>	<u>3.54</u>	<u>3.17</u>	<u>3.03</u>	28.28	19.62	17.71	<u>17.42</u>	<u>17.26</u>	19.02

- We conclude better covariance improves quality, diversity, likelihood
- We note full covariance infeasible; explore low-rank or blocks
- Table: we report FID across datasets; OCM competitive over steps

Thank You

Questions & Discussion